



BHARAT SINGH

Software Engineer



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Faridabad, India

[Portfolio Website \(Click\)](#)

PROFESSIONAL SUMMARY

Software Engineer with hands-on experience in Python, Flask, Qt (C++), and Computer Vision, currently working on the TCAS program under KAVACH, a safety-critical railway system. Strong background in automation, object detection, image segmentation, and OCR, with experience building desktop and web applications to improve testing efficiency and operational workflows.

WORK EXPERIENCE

Software Engineer

Quadrant Future Tek Ltd. – Hyderabad

Mar 2025 – Present

- Developing Qt (C++) desktop applications for system monitoring, and automation in railway safety systems.
- Building Flask-based applications and APIs to support backend automation and data processing.
- Designed and trained custom computer vision models for object detection and image segmentation to enable intelligent automation.
- Implemented OCR modules for automated data extraction and validation from images and documents.
- Contributed to automation tools that reduced manual effort in testing and validation of safety-critical workflows.

AIML Python Intern

SynapseIndia – Noida

Sep 2024 – Jan 2025

- Developed an advanced Retrieval-Augmented Generation (RAG) chatbot using LangChain and OpenAI.
- Engineered data ingestion pipelines for web scraping and document parsing using Python and ChromaDB.
- Built full-stack web applications using Django and React, integrating third-party APIs such as Stripe and Twilio.
- Implemented object detection models using OpenCV and integrated OCR for real-time document processing.
- Delivered scalable AI-driven solutions focused on automation and intelligent data retrieval.

TECHNICAL SKILLS

- **Programming Languages:** Python, C, Java (Core)
- **Frameworks & Libraries:** Qt (Qt Widgets), Flask, Django, OpenCV, LangChain
- **Computer Vision:** Object Detection, Image Segmentation, OCR
- **Machine Learning:** Custom ML/DL Models, Scikit-Learn
- **Data & APIs:** REST APIs, SQL, Pandas, NumPy
- **Tools & Platforms:** Git, Linux, Selenium, ChromaDB

EDUCATION

Bachelor of Technology (B.Tech) – Computer Science (AI & ML)

GLA University, Mathura CPI: 8.15

2021 – 2025

Intermediate -DAV Public School, Faridabad Score: 86.2%

2021

High School-DAV Public School, Faridabad Score: 87.88%

2019

SOFT SKILL

- Problem Solving
- Critical Thinking
- Team Collaboration
- Communication Skills

LANGUAGES

- English
- Hindi